

Gas Field Policy 23: Non-Lube Service

The following criteria must be met before a Siemens Energy compressor can be offered for non-lube service:

1. Discharge pressures are generally limited to 1500 PSIG.
2. Average piston speeds are not to exceed 750 ft/min.
3. Distance pieces are required.
4. Oil slingers must be provided on all piston rods. To accommodate these, distance pieces are required on the MOS and extra long distance pieces may be required on HOS/HOSS depending on the cylinder size and/or pressure rating.
5. Packing cases must be water cooled when discharge pressures exceed 250 psig.
6. Cylinder bores must have a 16 RMS special finish and be sprayed with Teflon.
7. Piston rods to be coated only when application absolutely requires it.
8. There may be special requirements beyond what is normally required for lubed applications such as cylinder liners, special pistons, distance pieces, etc.
9. Estimated actual discharge gas temperature is not to exceed 250° F for gas cooled cylinders and 300° F for water cooled cylinders.
10. Process vessels and piping must be carefully designed to prevent liquids and foreign debris from entering the compressor. They must be very carefully cleaned prior to mounting on the compressor packaged unit. Siemens Energy also recommends that a coating such as Hunting Industrial #8101 (or equal) be applied to all vessels and gas piping to help maintain cleanliness. Foreign debris from the vessels or piping along with liquids can form a greasy paste that has the potential to block the venting of piston and packing rings and/or adhere to the cylinder bore and valves with a detrimental effect on service life.
11. Abrasive fines from such things as molecular sieves or desiccant dryers, glass or ceramics from towers or columns, etc. must be eliminated or at least filtered to below 1 micron.
12. Siemens Energy will provide the non-lubricated compressor cylinders and packing cases drilled for lubricated service and plugged. In the event that the Process Gas System requirements cannot be achieved to provide acceptable service life, this will facilitate conversion to lubricated construction and operation if necessary.
13. Wear parts in non-lubricated compressor cylinders typically experience shorter service life than do wear parts in lubricated compressor cylinders which translates into more frequent maintenance stoppages. The user should ensure they have an adequate supply of replacement parts on hand to avoid unnecessary down time.
14. Start-up procedures must be submitted to Siemens Energy for review.
15. On HOS & HOSS compressors, 26" - 28" – "clamshell type" – cylinders cannot be used.
16. Stroke should be limited to 5" or 6". Should a 7" stroke be required for an application, this will require engineering review for any piston design changes.